SCV

AI Drone Tracking System

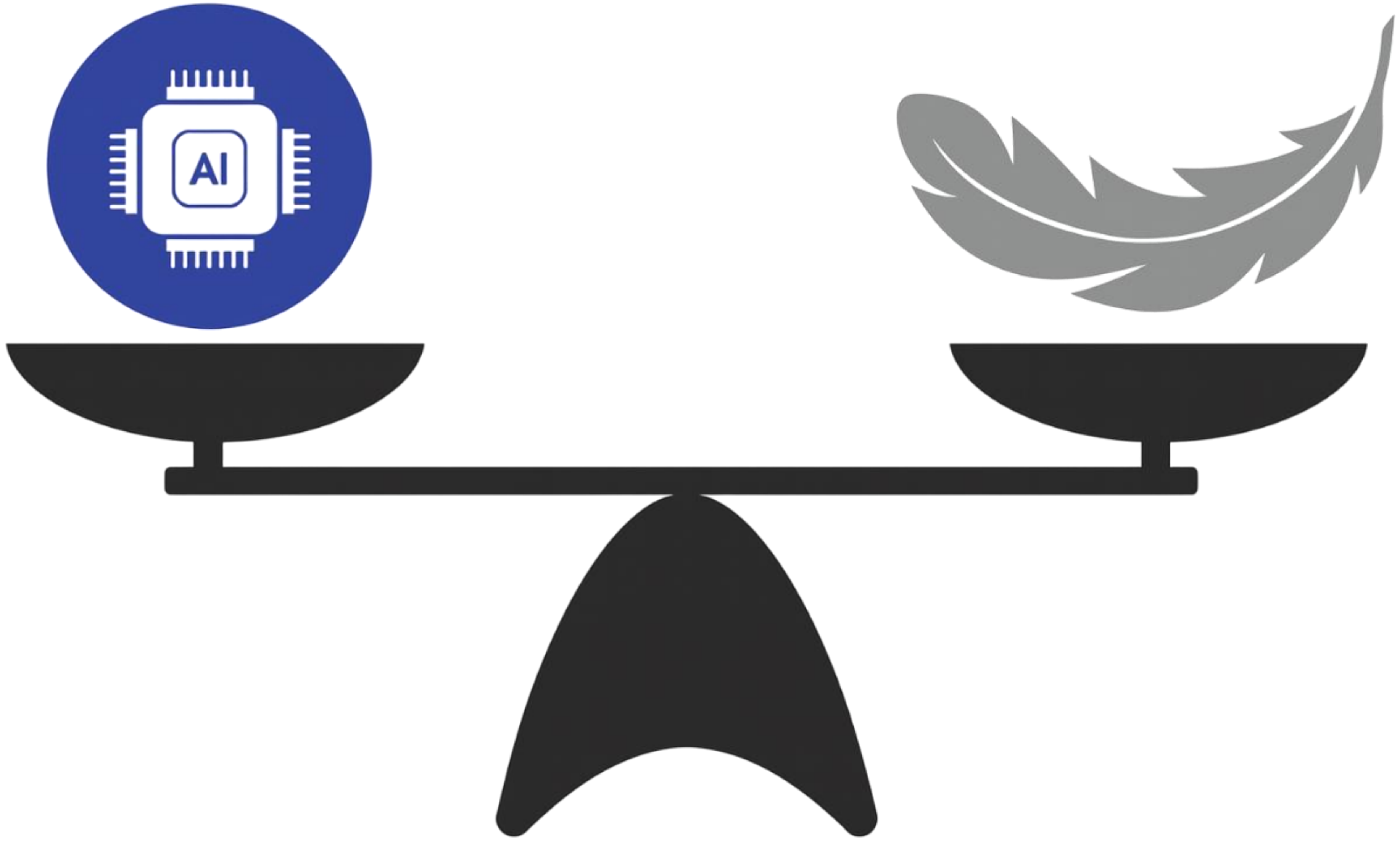
2조

목차

01 배경

01 목표

경량화된 드론 시스템 구현



02 팀소개

황진영 | Leader

- PM
- GUI

김국현 | AI

- Hailo Compile
- AI 최적화

정태윤 | Firmware

- 팬틸트 카메라
- Yocto

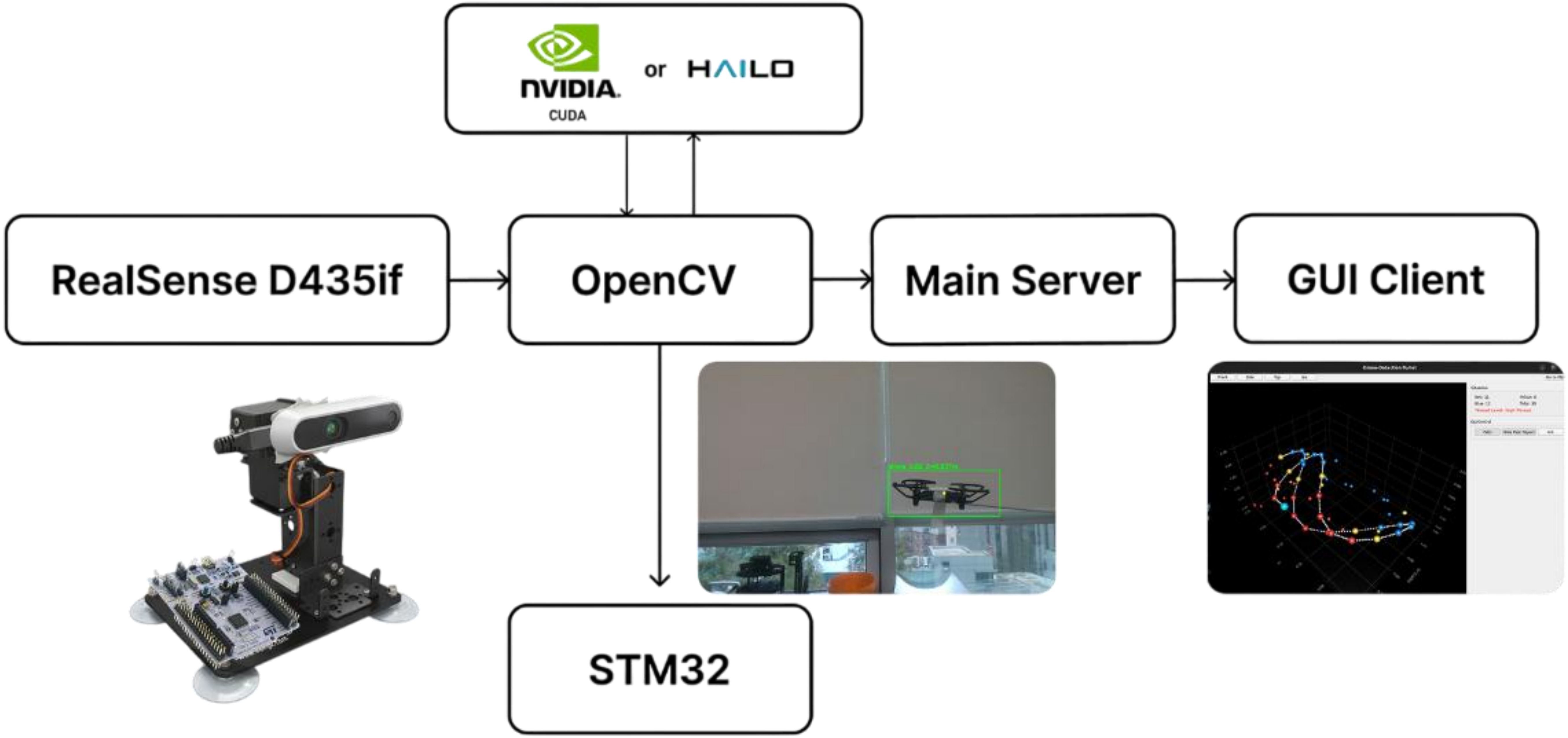
윤동준 | OPEN CV

- 영상 처리
- 카메라 스트리밍

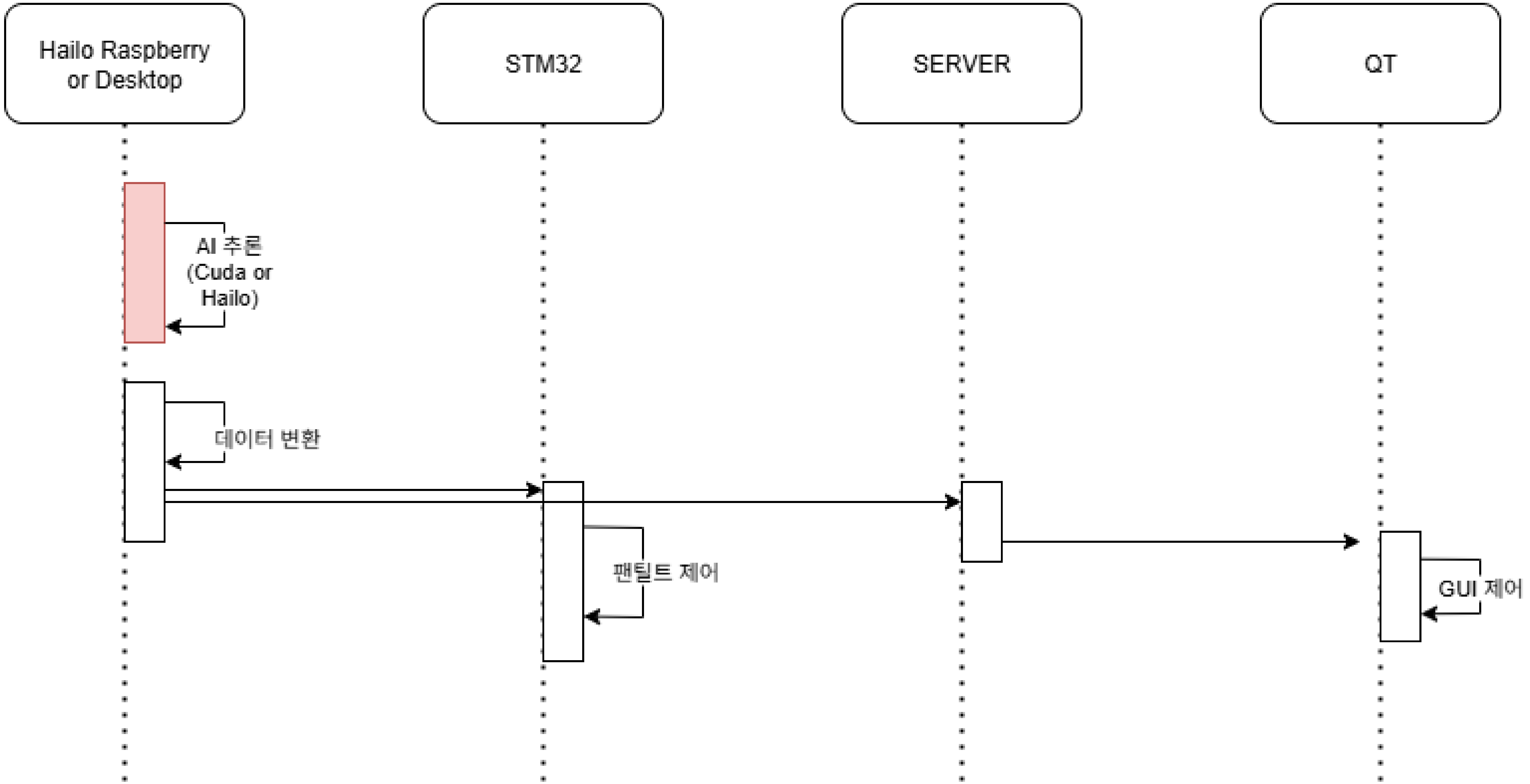
박성준 | AI

- Hailo Application

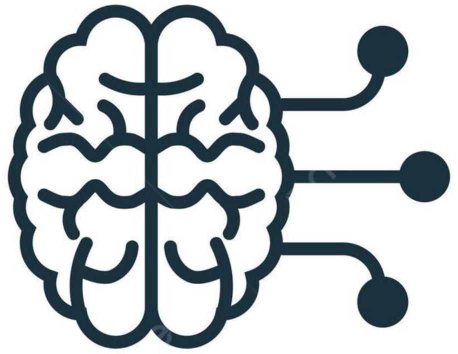
03 구조도


NVIDIA
CUDA
or HAILO
RealSense D435if
OpenCV
Main Server
GUI Client
STM32

04 구현

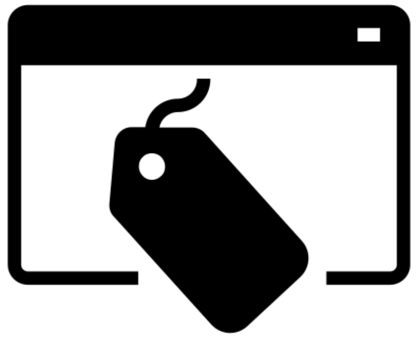

Hailo Raspberry or Desktop
STM32
SERVER
QT
AI 추론 (Cuda or Hailo)
데이터 변환
팬틸트 제어
GUI 제어

04 구현 - AI



YOLOv8s

- Yolov8 : 비전 작업에 유리
- 's' 모델 : 엣지 디바이스에서 유리



Label 1개

- 클래스 혼동 감소로 정확도 향상
- 적은 데이터로 높은 성능 기대

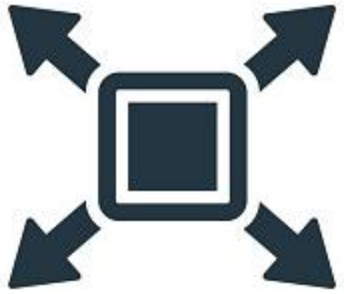
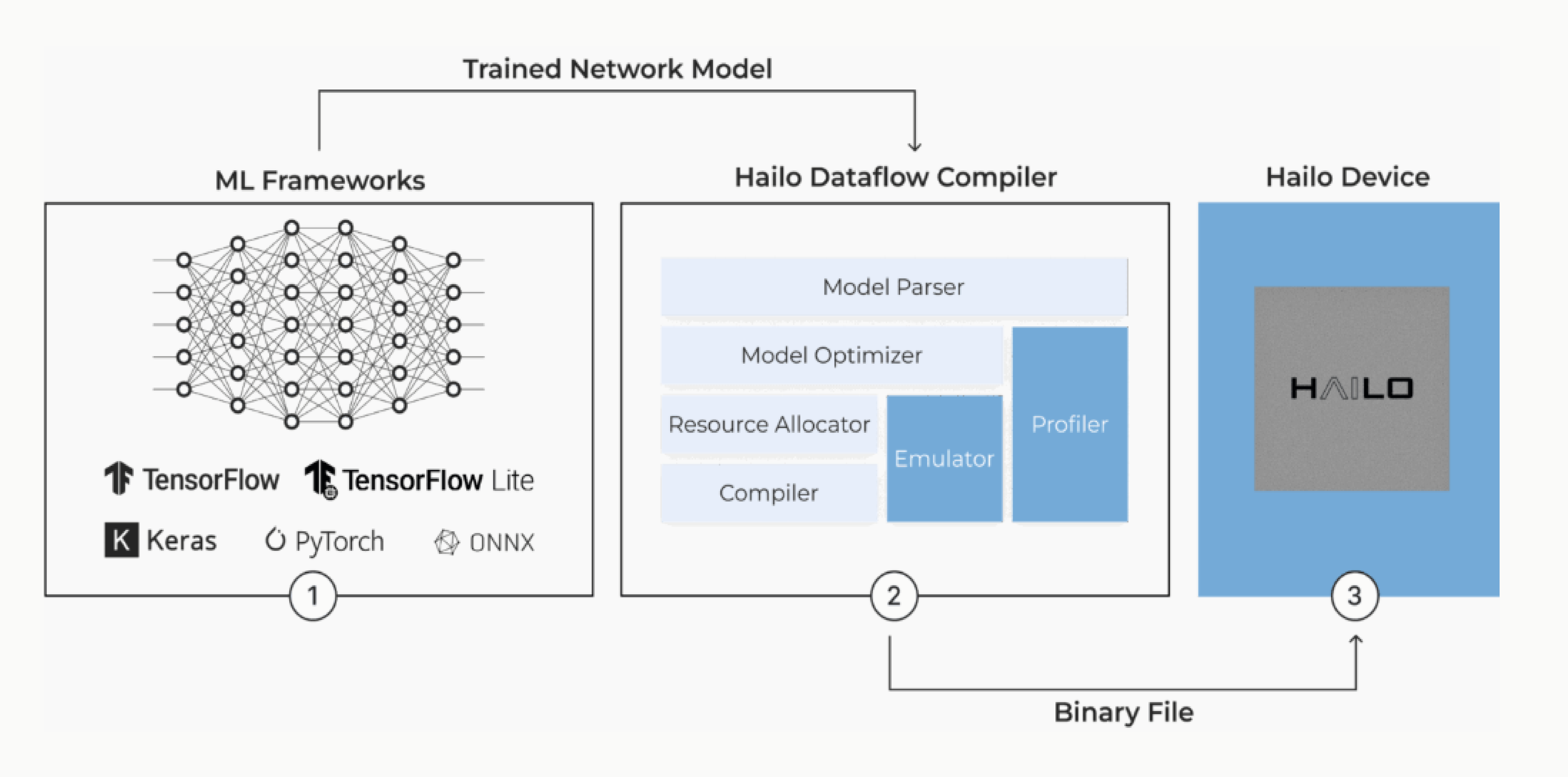


Image Size

- 이미지 축소시 추론 속도 향상
- 640x640을 320x320으로 변경

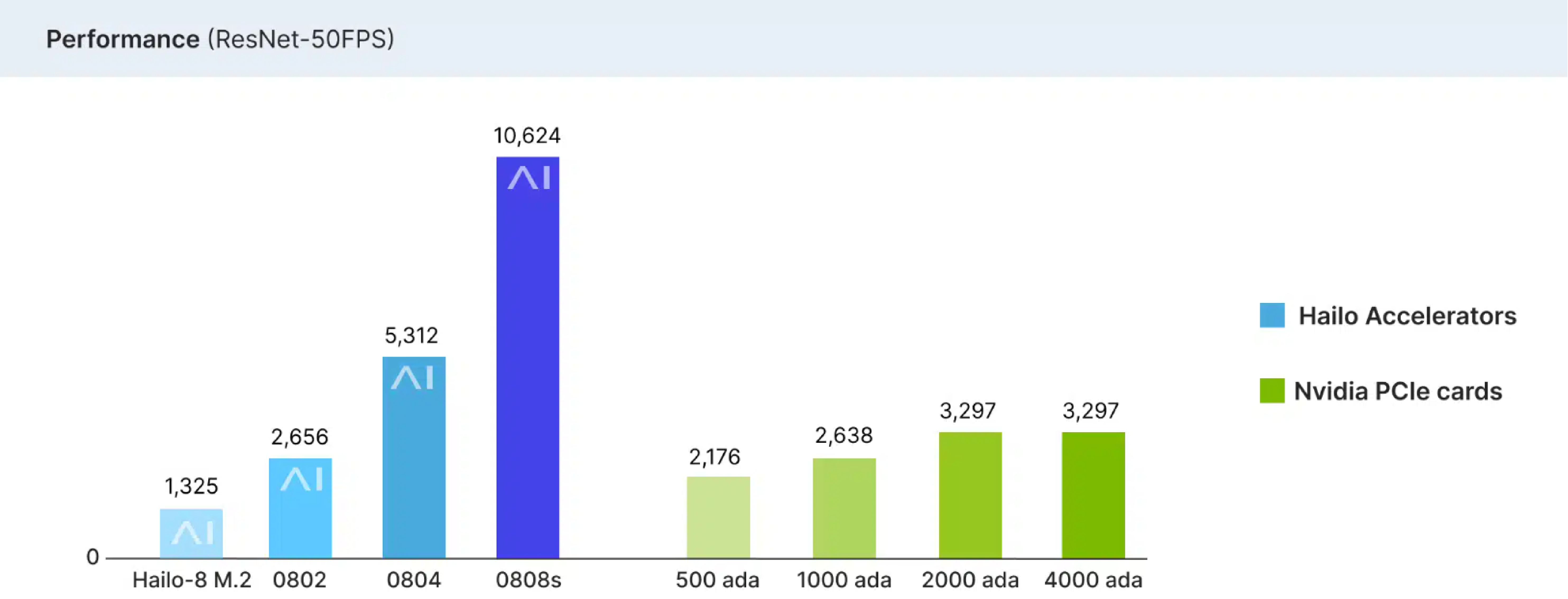
04 구현 - AI



Trained Network Model
ML Frameworks
Hailo Dataflow Compiler
Hailo Device
TensorFlow
TensorFlow Lite
Keras
PyTorch
ONNX
Model Parser
Model Optimizer
Resource Allocator
Compiler
Emulator
Profiler
HAILO
1
2
3
Binary File

04 구현 – AI

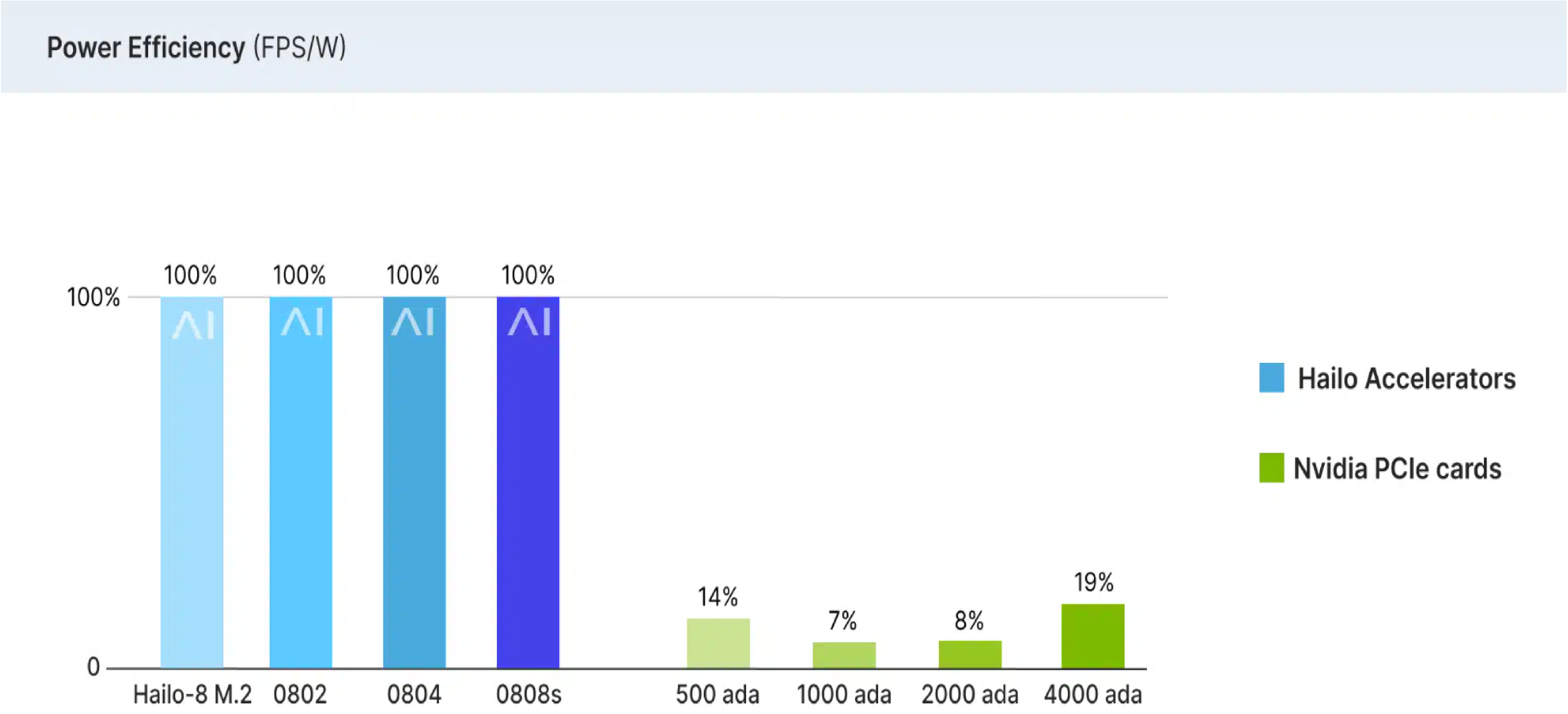
WHY?



Performance (ResNet-50FPS)
10,624
5,312
2,656
1,325
2,176
2,638
3,297
3,297
0
Hailo-8 M.2
0802
0804
0808s
500 ada
1000 ada
2000 ada
4000 ada
Hailo Accelerators
Nvidia PCIe cards

04 구현 – AI

WHY?



Power Efficiency (FPS/W)
100%
0
100%
100%
100%
100%
14%
7%
8%
19%
Hailo-8 M.2
0802
0804
0808s
500 ada
1000 ada
2000 ada
4000 ada
Hailo Accelerators
Nvidia PCIe cards

04 구현 – AI

Cuda with NVIDIA GPU

1.좋은 성능과 병렬 처리 능력

2.방대한 라이브러리로 인한 빠른 구성 가능

3.OpenCV와의 좋은 연동성

04 구현 – AI

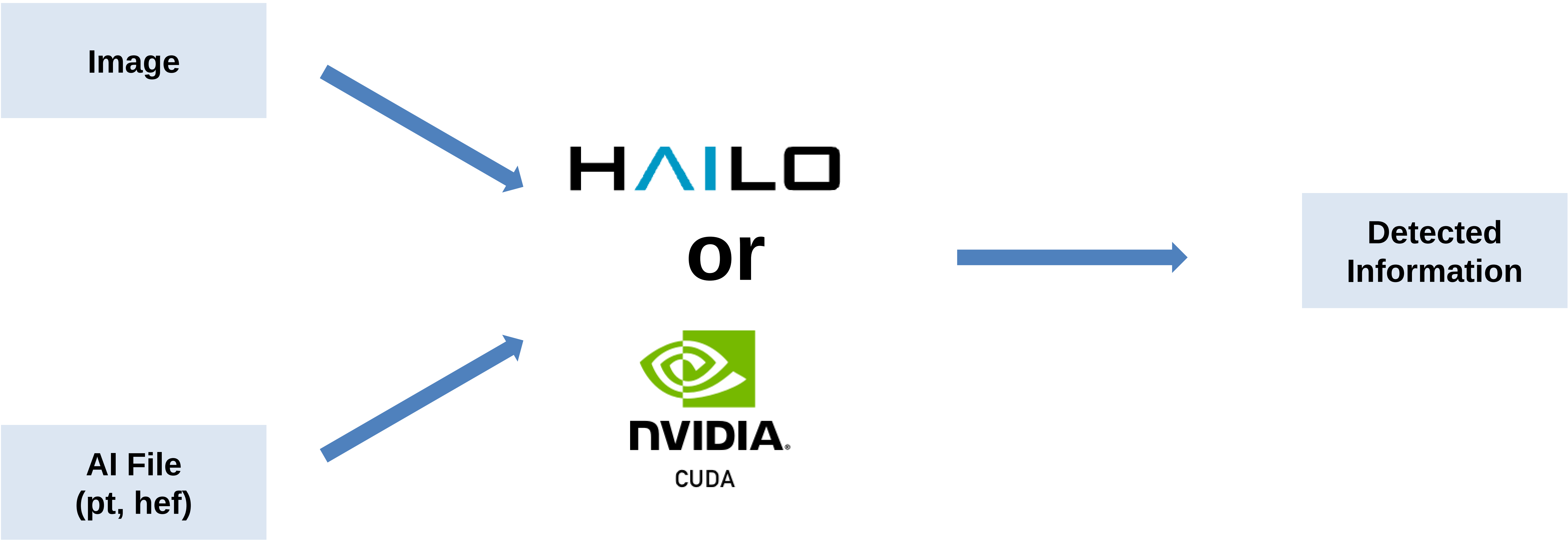
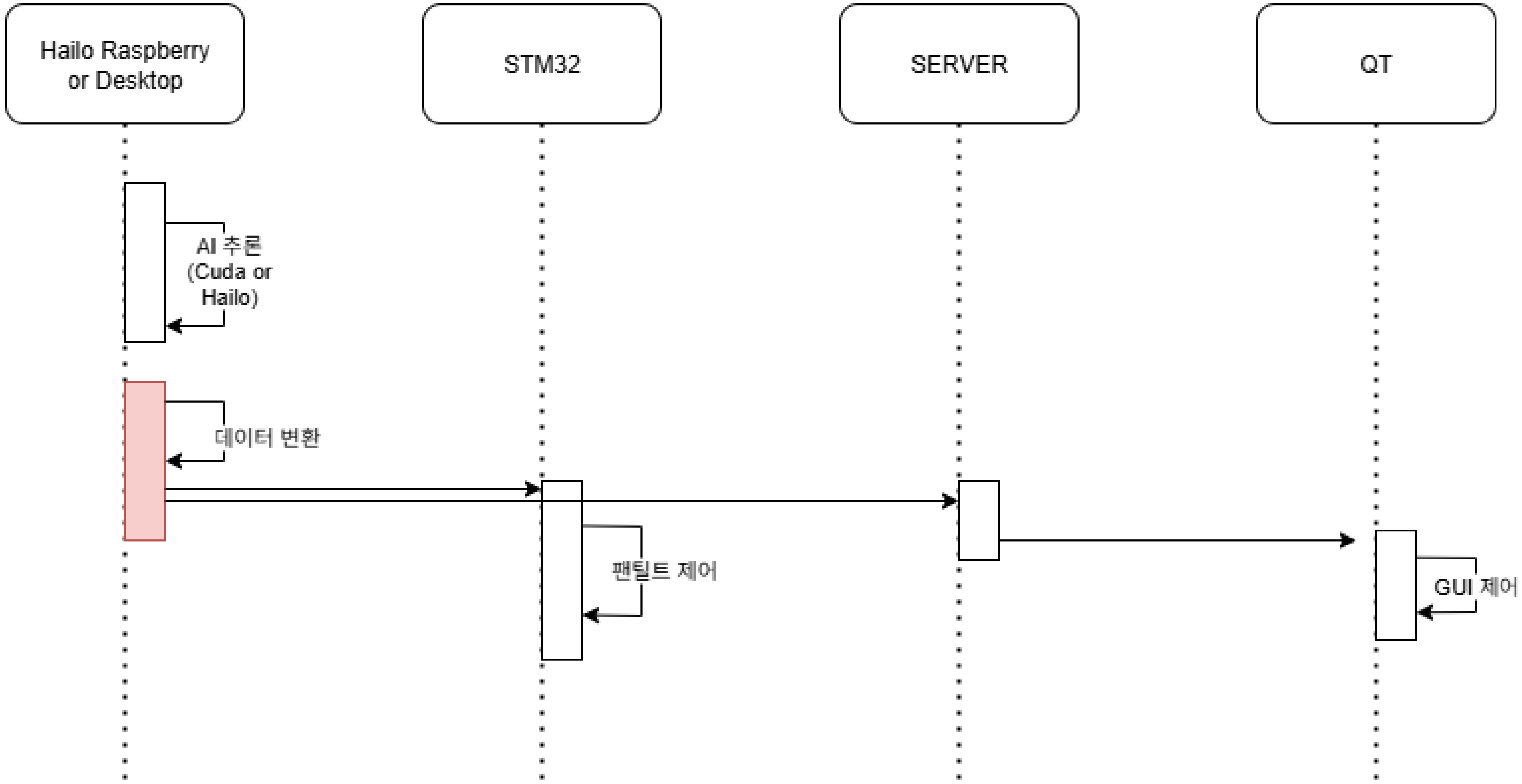
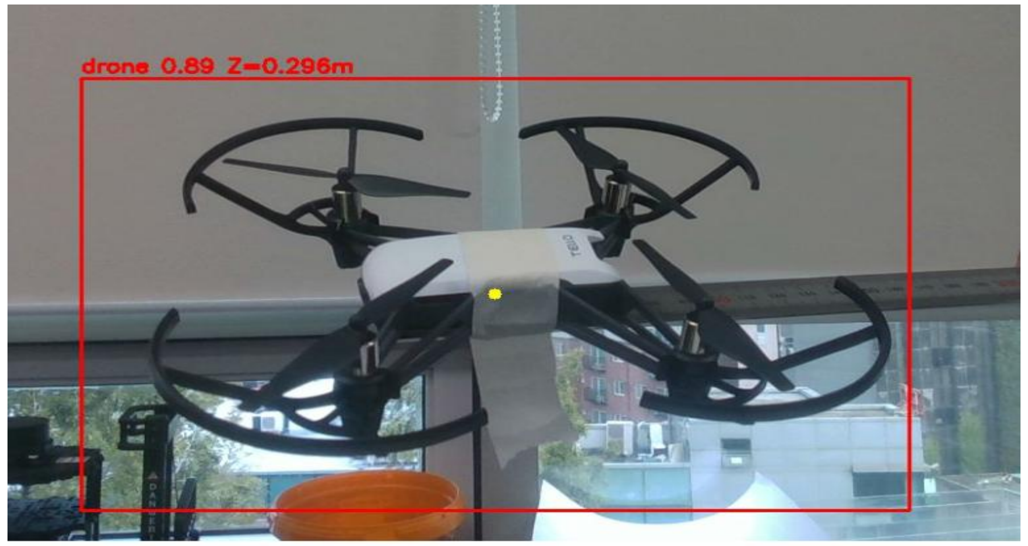


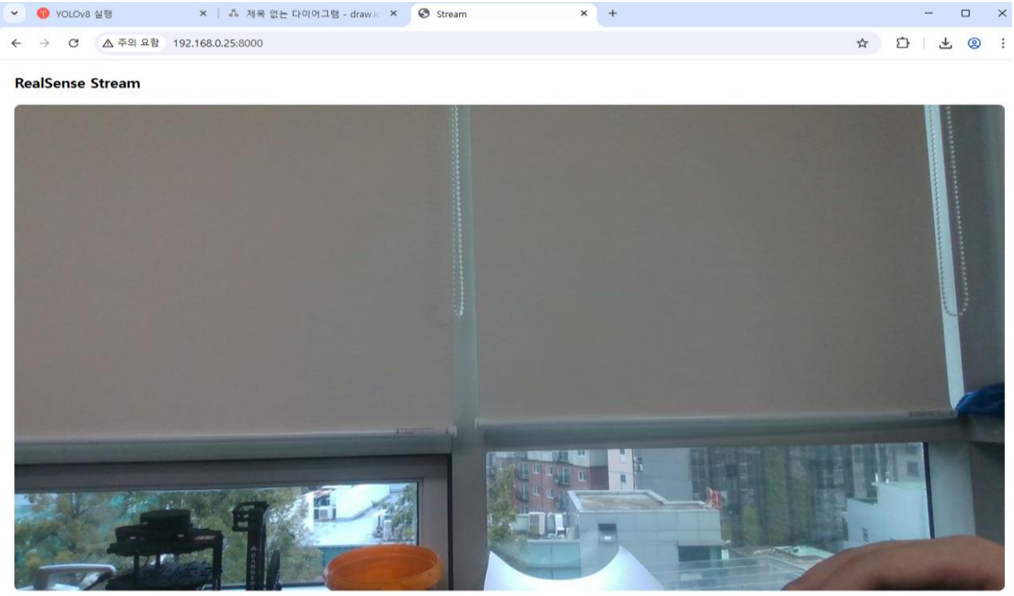
Image
AI File
(pt, hef)
HAILO
or
NVIDIA
CUDA
Detected
Information

04 구현


Hailo Raspberry or Desktop
STM32
SERVER
QT
AI 추론 (Cuda or Hailo)
데이터 변환
팬틸트 제어
GUI 제어

04 구현



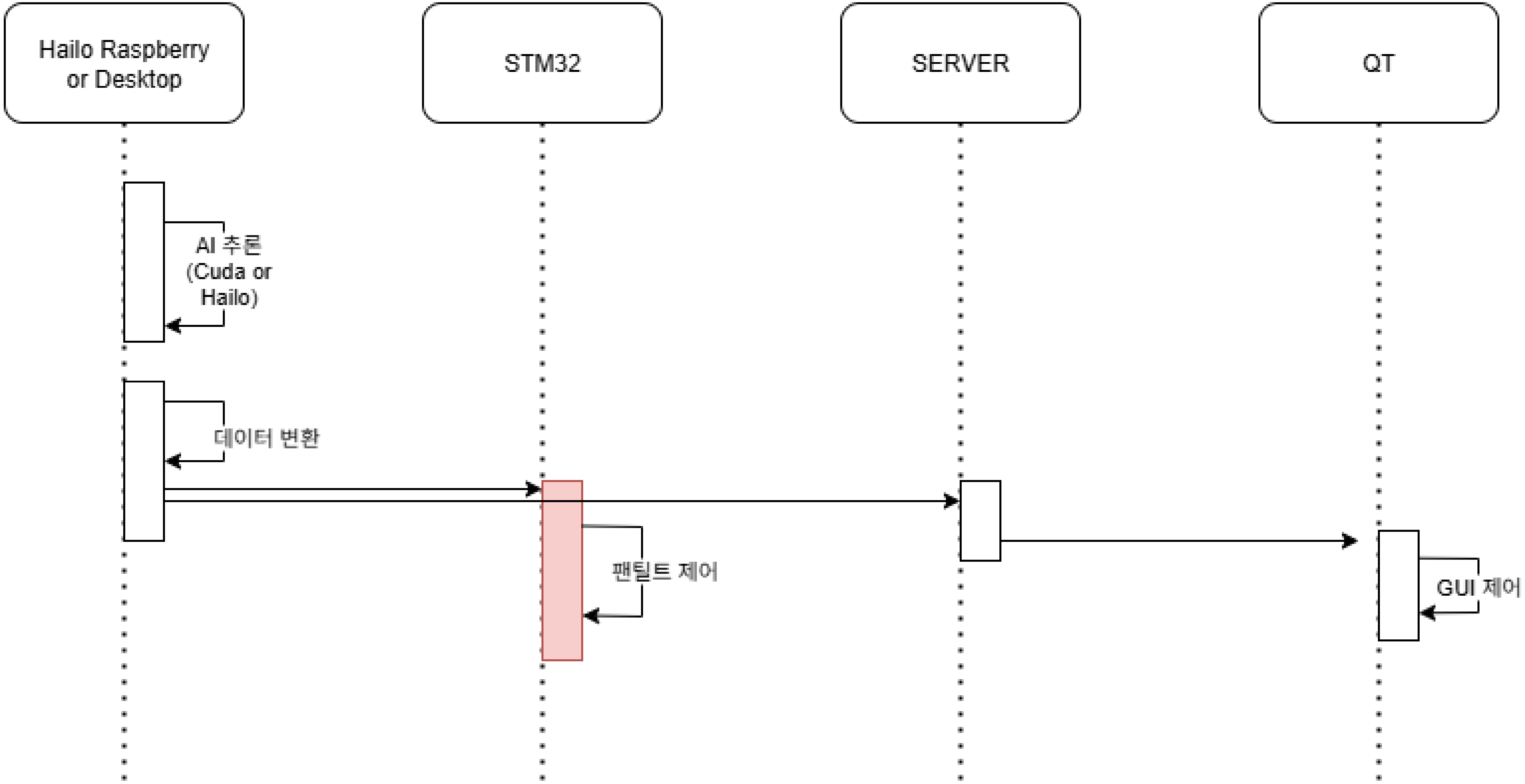


1. 거리기반 위험도 정의

2. 영상 스트리밍

3. Detecting 결과 데이터 송수신

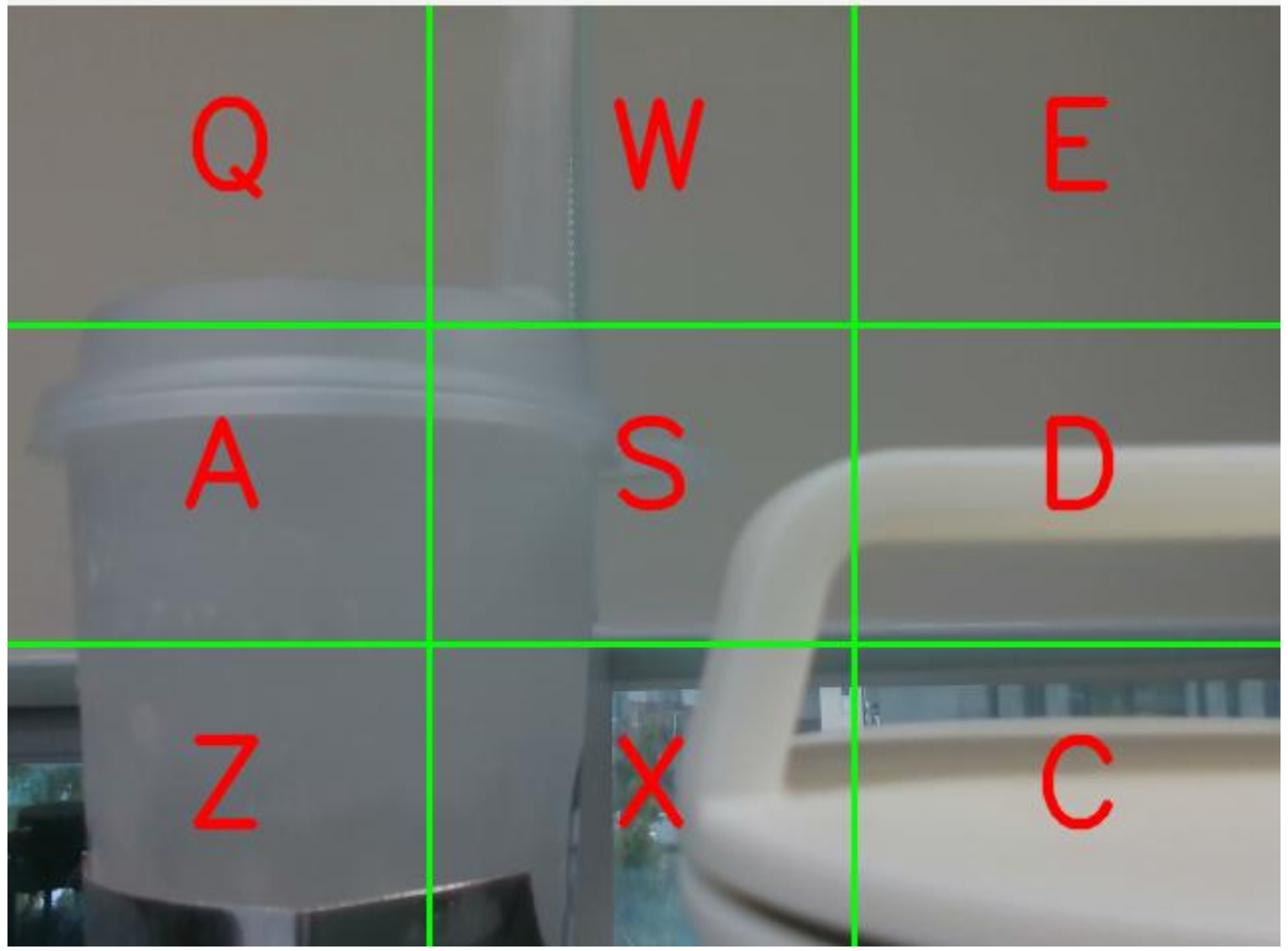
04 구현

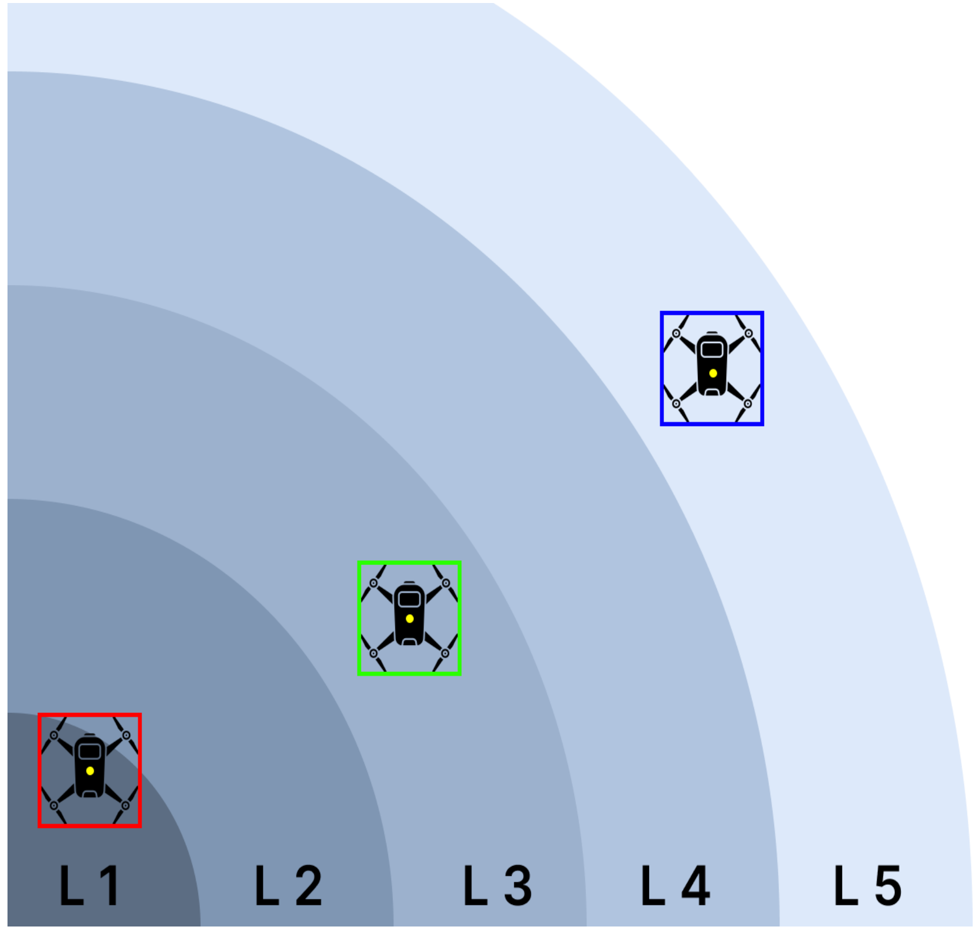


Hailo Raspberry or Desktop
STM32
SERVER
QT
AI 추론 (Cuda or Hailo)
데이터 변환
팬틸트 제어
GUI 제어

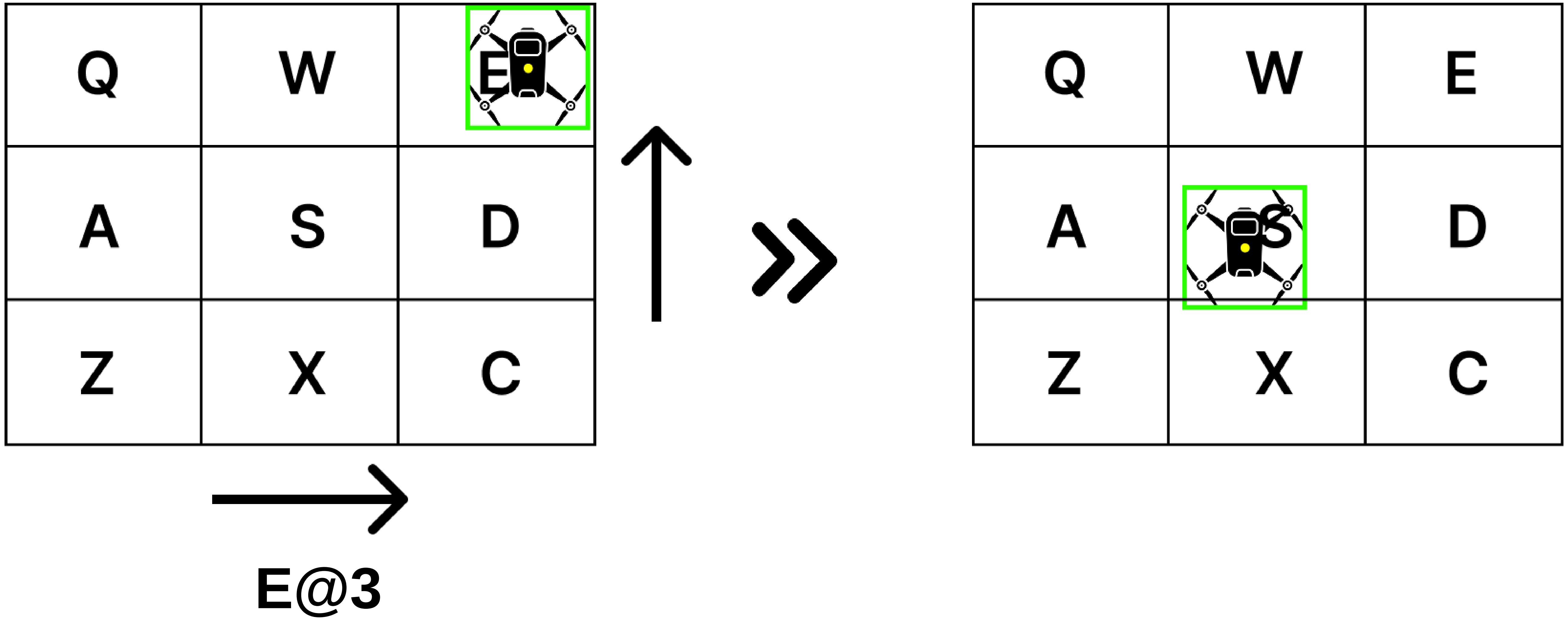
04 구현 - Pan-Tilt

STM32 UART 통신 : Q@1

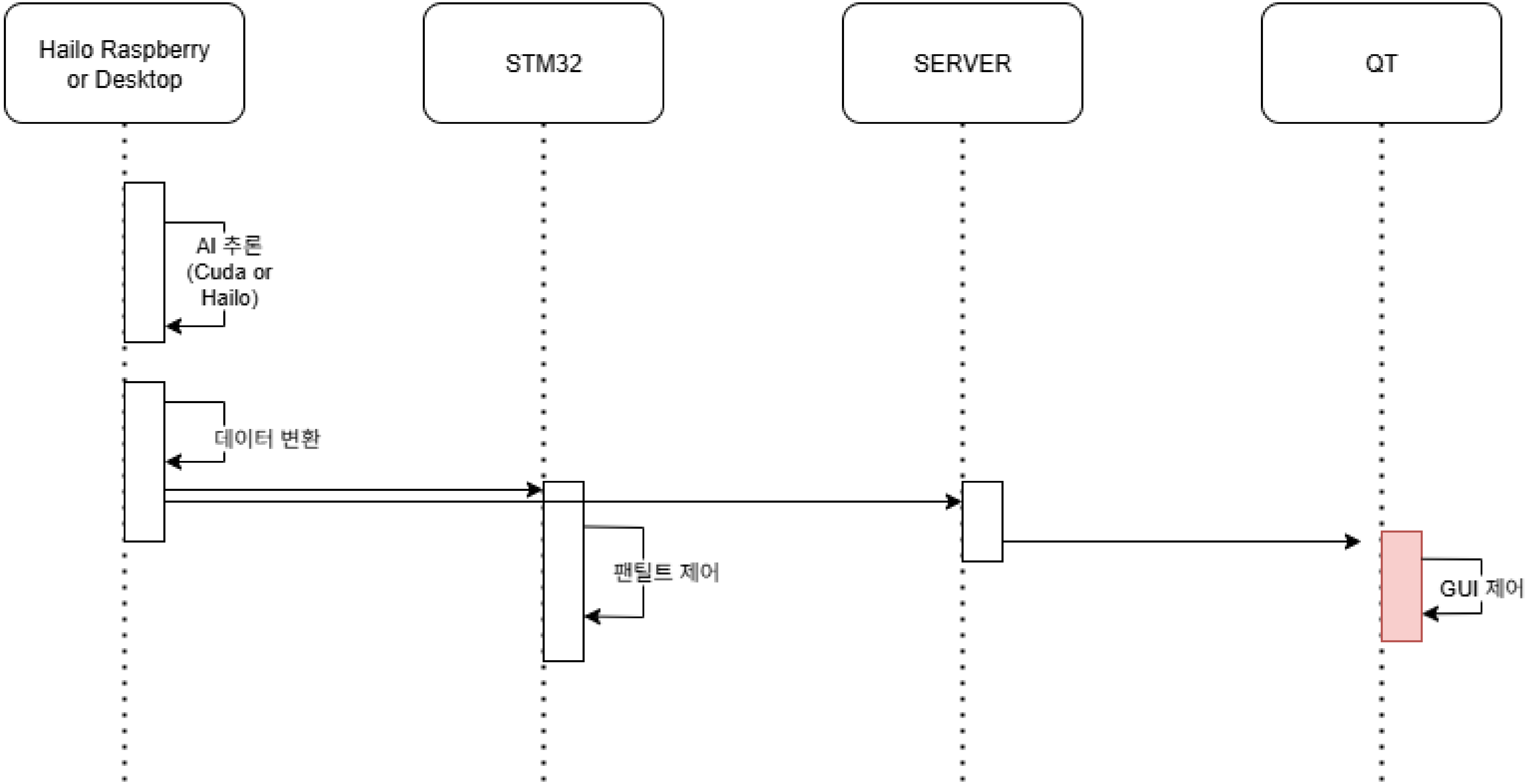




04 구현 - Pan-Tilt



04 구현


Hailo Raspberry or Desktop
STM32
SERVER
QT
AI 추론 (Cuda or Hailo)
데이터 변환
팬틸트 제어
GUI 제어

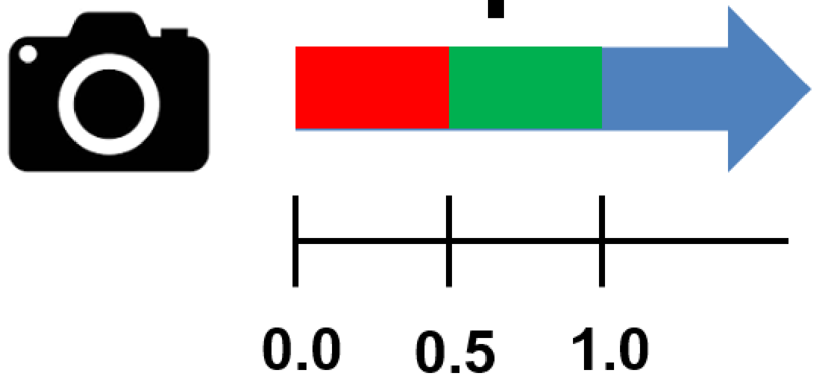
04 구현 – GUI

QT : X@Y@dept@label

Detection 결과의 중점의 좌표

드론의 거리

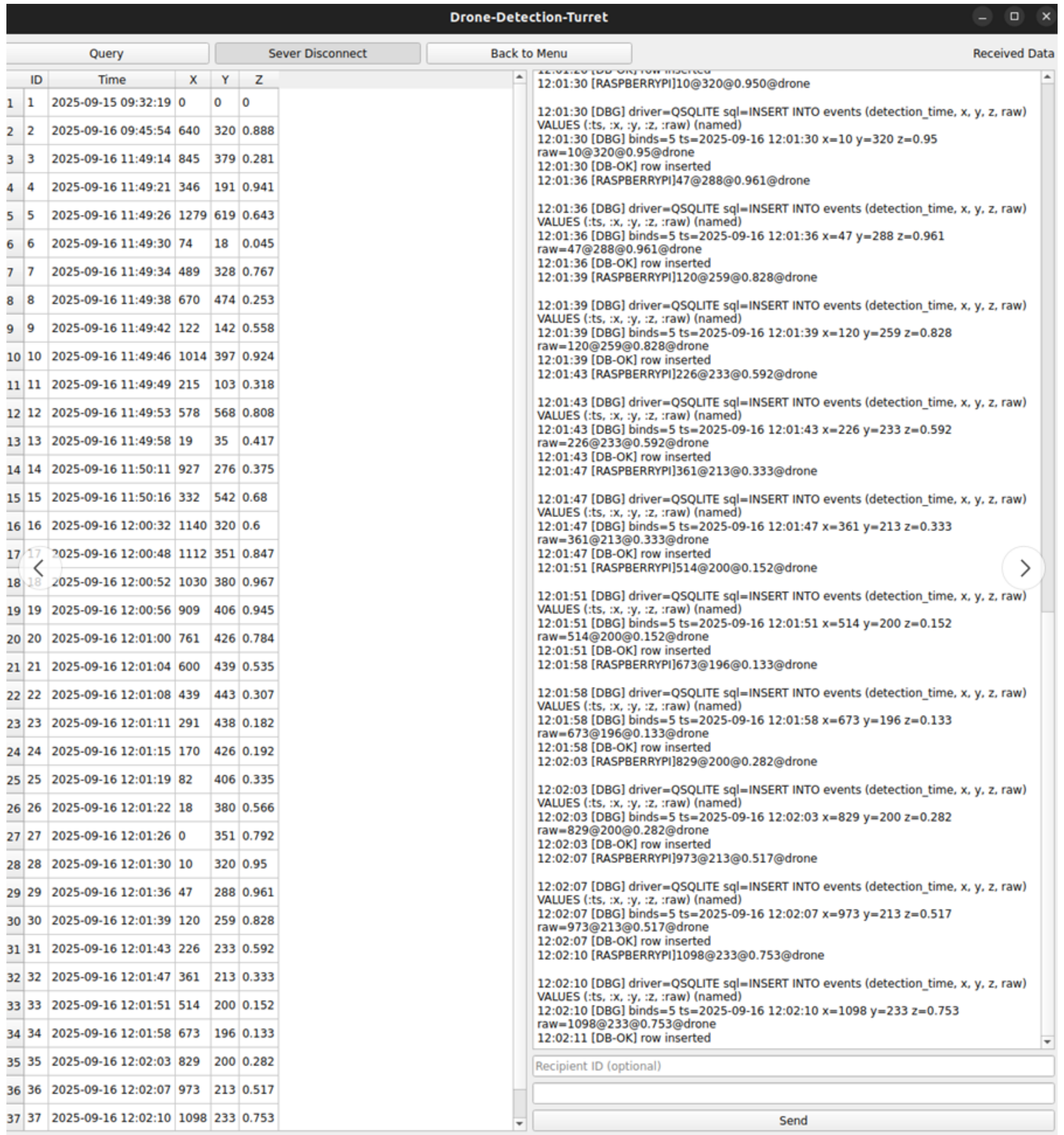
라벨명

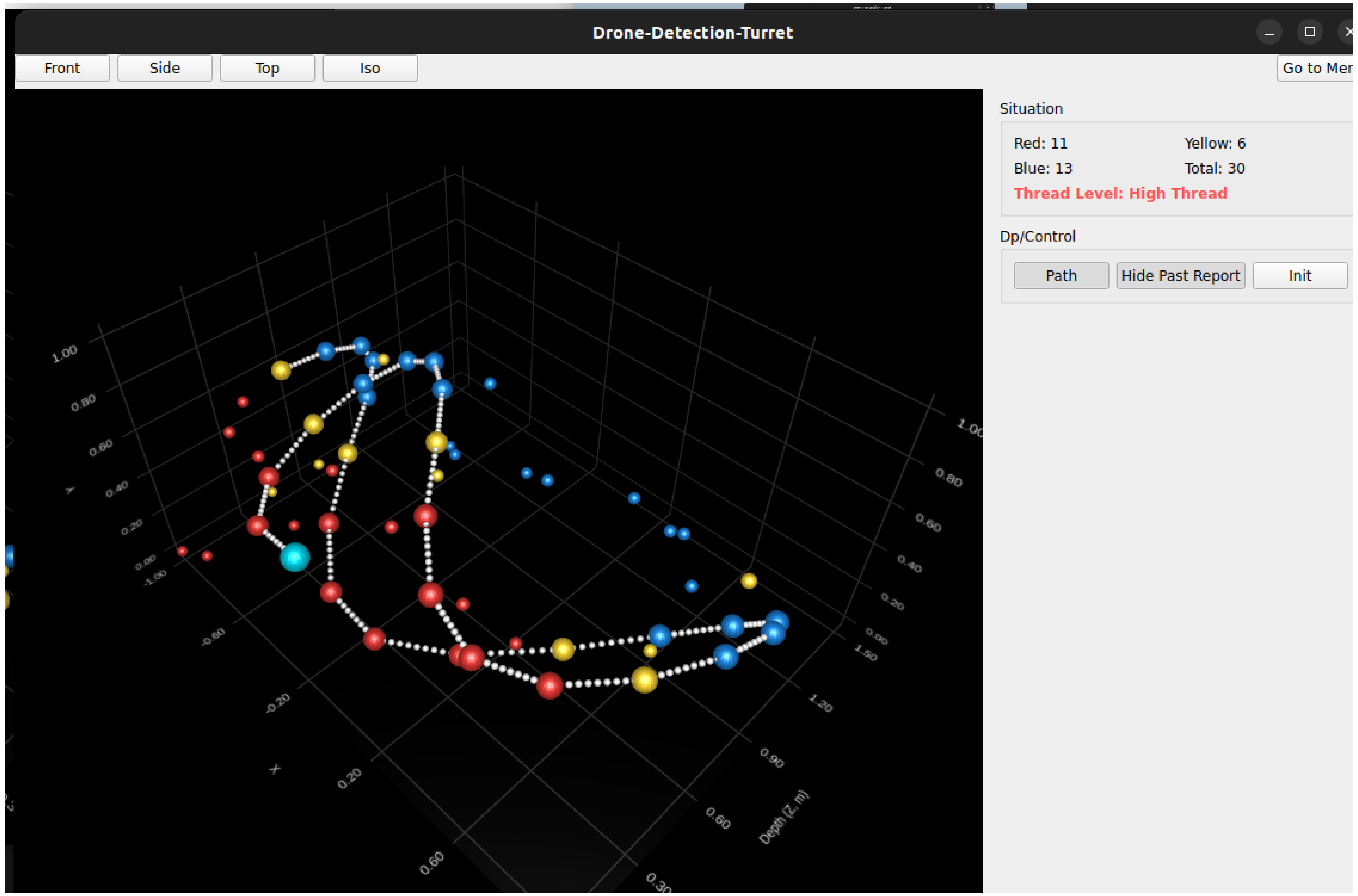


04 구현 – GUI

Custom Linux for GUI

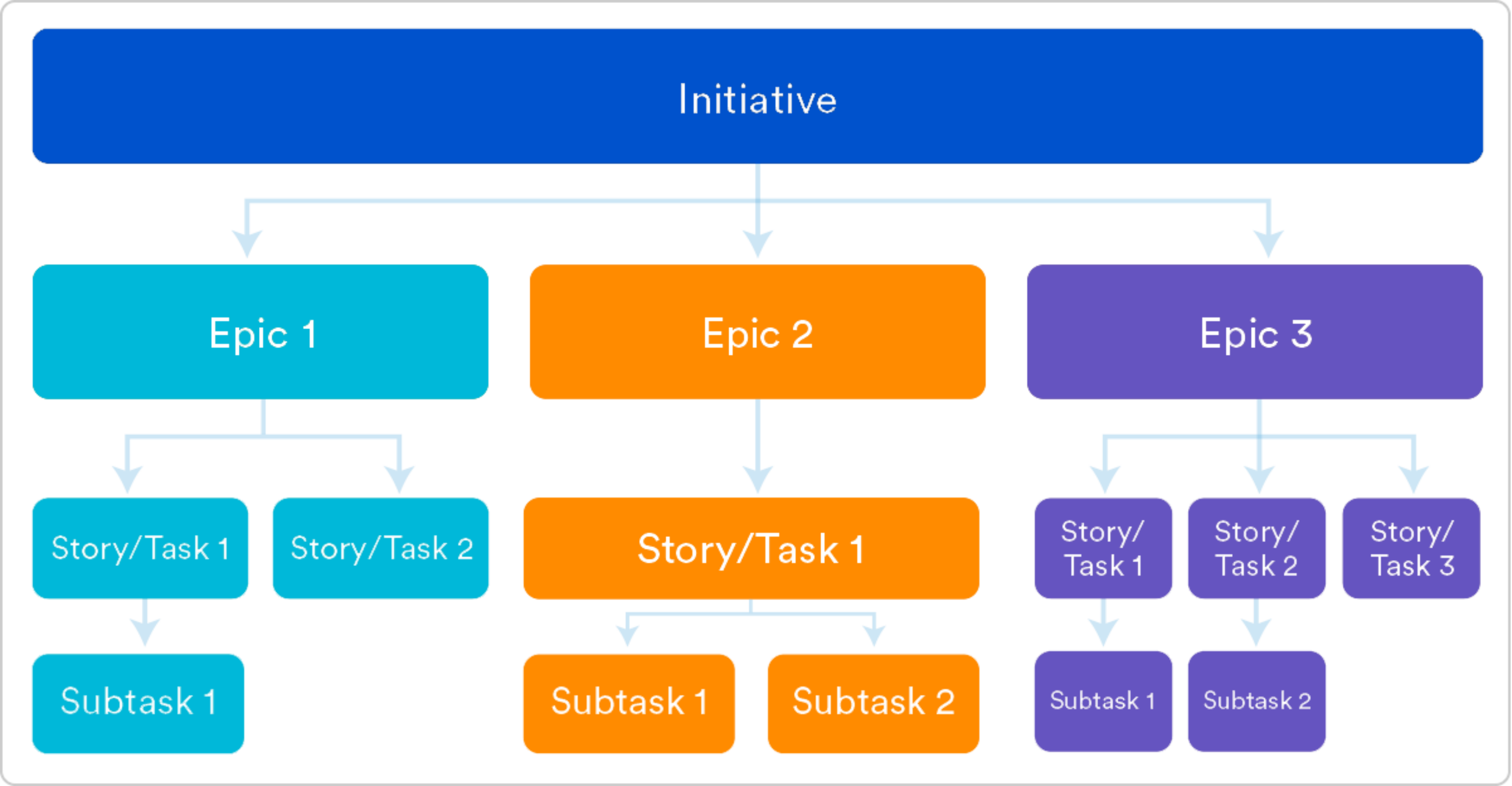
04 구현 – GUI

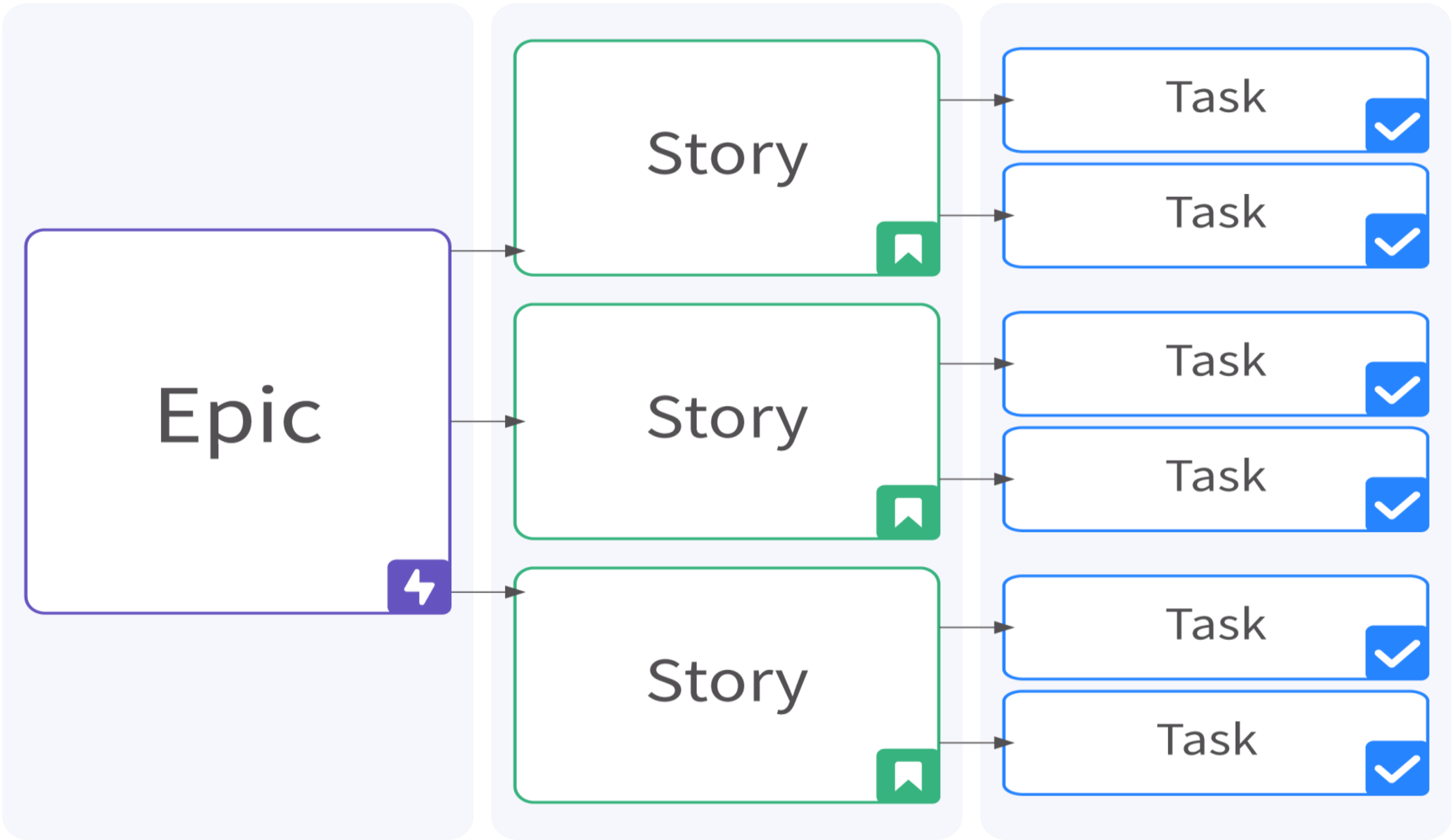

Drone-Detection-Turret
Query
Sever Disconnect
Back to Menu
Received Data
ID Time X Y Z
1 1 2025-09-15 09:32:19 0 0 0
2 2 2025-09-16 09:45:54 640 320 0.888
3 3 2025-09-16 11:49:14 845 379 0.281
4 4 2025-09-16 11:49:21 346 191 0.941
5 5 2025-09-16 11:49:26 1279 619 0.643
6 6 2025-09-16 11:49:30 74 18 0.045
7 7 2025-09-16 11:49:34 489 328 0.767
8 8 2025-09-16 11:49:38 670 474 0.253
9 9 2025-09-16 11:49:42 122 142 0.558
10 10 2025-09-16 11:49:46 1014 397 0.924
11 11 2025-09-16 11:49:49 215 103 0.318
12 12 2025-09-16 11:49:53 578 568 0.808
13 13 2025-09-16 11:49:58 19 35 0.417
14 14 2025-09-16 11:50:11 927 276 0.375
15 15 2025-09-16 11:50:16 332 542 0.68
16 16 2025-09-16 12:00:32 1140 320 0.6
17 17 2025-09-16 12:00:48 1112 351 0.847
18 18 2025-09-16 12:00:52 1030 380 0.967
19 19 2025-09-16 12:00:56 909 406 0.945
20 20 2025-09-16 12:01:00 761 426 0.784
21 21 2025-09-16 12:01:04 600 439 0.535
22 22 2025-09-16 12:01:08 439 443 0.307
23 23 2025-09-16 12:01:11 291 438 0.182
24 24 2025-09-16 12:01:15 170 426 0.192
25 25 2025-09-16 12:01:19 82 406 0.335
26 26 2025-09-16 12:01:22 18 380 0.566
27 27 2025-09-16 12:01:26 0 351 0.792
28 28 2025-09-16 12:01:30 10 320 0.95
29 29 2025-09-16 12:01:36 47 288 0.961
30 30 2025-09-16 12:01:39 120 259 0.828
31 31 2025-09-16 12:01:43 226 233 0.592
32 32 2025-09-16 12:01:47 361 213 0.333
33 33 2025-09-16 12:01:51 514 200 0.152
34 34 2025-09-16 12:01:58 673 196 0.133
35 35 2025-09-16 12:02:03 829 200 0.282
36 36 2025-09-16 12:02:07 973 213 0.517
37 37 2025-09-16 12:02:10 1098 233 0.753
12:01:30 [RASPBERRYPI]10@320@0.950@drone
12:01:30 [DBG] driver=QSQLITE sql=INSERT INTO events (detection_time, x, y, z, raw) VALUES (:ts, :x, :y, :z, :raw) (named)
12:01:30 [DBG] binds=5 ts=2025-09-16 12:01:30 x=10 y=320 z=0.95 raw=10@320@0.95@drone
12:01:30 [DB-OK] row inserted
12:01:36 [RASPBERRYPI]47@288@0.961@drone
12:01:36 [DBG] driver=QSQLITE sql=INSERT INTO events (detection_time, x, y, z, raw) VALUES (:ts, :x, :y, :z, :raw) (named)
12:01:36 [DBG] binds=5 ts=2025-09-16 12:01:36 x=47 y=288 z=0.961 raw=47@288@0.961@drone
12:01:36 [DB-OK] row inserted
12:01:39 [RASPBERRYPI]120@259@0.828@drone
12:01:39 [DBG] driver=QSQLITE sql=INSERT INTO events (detection_time, x, y, z, raw) VALUES (:ts, :x, :y, :z, :raw) (named)
12:01:39 [DBG] binds=5 ts=2025-09-16 12:01:39 x=120 y=259 z=0.828 raw=120@259@0.828@drone
12:01:39 [DB-OK] row inserted
12:01:43 [RASPBERRYPI]226@233@0.592@drone
12:01:43 [DBG] driver=QSQLITE sql=INSERT INTO events (detection_time, x, y, z, raw) VALUES (:ts, :x, :y, :z, :raw) (named)
12:01:43 [DBG] binds=5 ts=2025-09-16 12:01:43 x=226 y=233 z=0.592 raw=226@233@0.592@drone
12:01:43 [DB-OK] row inserted
12:01:47 [RASPBERRYPI]361@213@0.333@drone
12:01:47 [DBG] driver=QSQLITE sql=INSERT INTO events (detection_time, x, y, z, raw) VALUES (:ts, :x, :y, :z, :raw) (named)
12:01:47 [DBG] binds=5 ts=2025-09-16 12:01:47 x=361 y=213 z=0.333 raw=361@213@0.333@drone
12:01:47 [DB-OK] row inserted
12:01:51 [RASPBERRYPI]514@200@0.152@drone
12:01:51 [DBG] driver=QSQLITE sql=INSERT INTO events (detection_time, x, y, z, raw) VALUES (:ts, :x, :y, :z, :raw) (named)
12:01:51 [DBG] binds=5 ts=2025-09-16 12:01:51 x=514 y=200 z=0.152 raw=514@200@0.152@drone
12:01:51 [DB-OK] row inserted
12:01:58 [RASPBERRYPI]673@196@0.133@drone
12:01:58 [DBG] driver=QSQLITE sql=INSERT INTO events (detection_time, x, y, z, raw) VALUES (:ts, :x, :y, :z, :raw) (named)
12:01:58 [DBG] binds=5 ts=2025-09-16 12:01:58 x=673 y=196 z=0.133 raw=673@196@0.133@drone
12:01:58 [DB-OK] row inserted
12:02:03 [RASPBERRYPI]829@200@0.282@drone
12:02:03 [DBG] driver=QSQLITE sql=INSERT INTO events (detection_time, x, y, z, raw) VALUES (:ts, :x, :y, :z, :raw) (named)
12:02:03 [DBG] binds=5 ts=2025-09-16 12:02:03 x=829 y=200 z=0.282 raw=829@200@0.282@drone
12:02:03 [DB-OK] row inserted
12:02:07 [RASPBERRYPI]973@213@0.517@drone
12:02:07 [DBG] driver=QSQLITE sql=INSERT INTO events (detection_time, x, y, z, raw) VALUES (:ts, :x, :y, :z, :raw) (named)
12:02:07 [DBG] binds=5 ts=2025-09-16 12:02:07 x=973 y=213 z=0.517 raw=973@213@0.517@drone
12:02:07 [DB-OK] row inserted
12:02:10 [RASPBERRYPI]1098@233@0.753@drone
12:02:10 [DBG] driver=QSQLITE sql=INSERT INTO events (detection_time, x, y, z, raw) VALUES (:ts, :x, :y, :z, :raw) (named)
12:02:10 [DBG] binds=5 ts=2025-09-16 12:02:10 x=1098 y=233 z=0.753 raw=1098@233@0.753@drone
12:02:11 [DB-OK] row inserted
Recipient ID (optional)
Send


Drone-Detection-Turret
Front
Side
Top
Iso
Go to Mer
Situation
Red: 11
Yellow: 6
Blue: 13
Total: 30
Thread Level: High Thread
Dp/Control
Path
Hide Past Report
Init
Depth (Z, m)

05 이슈

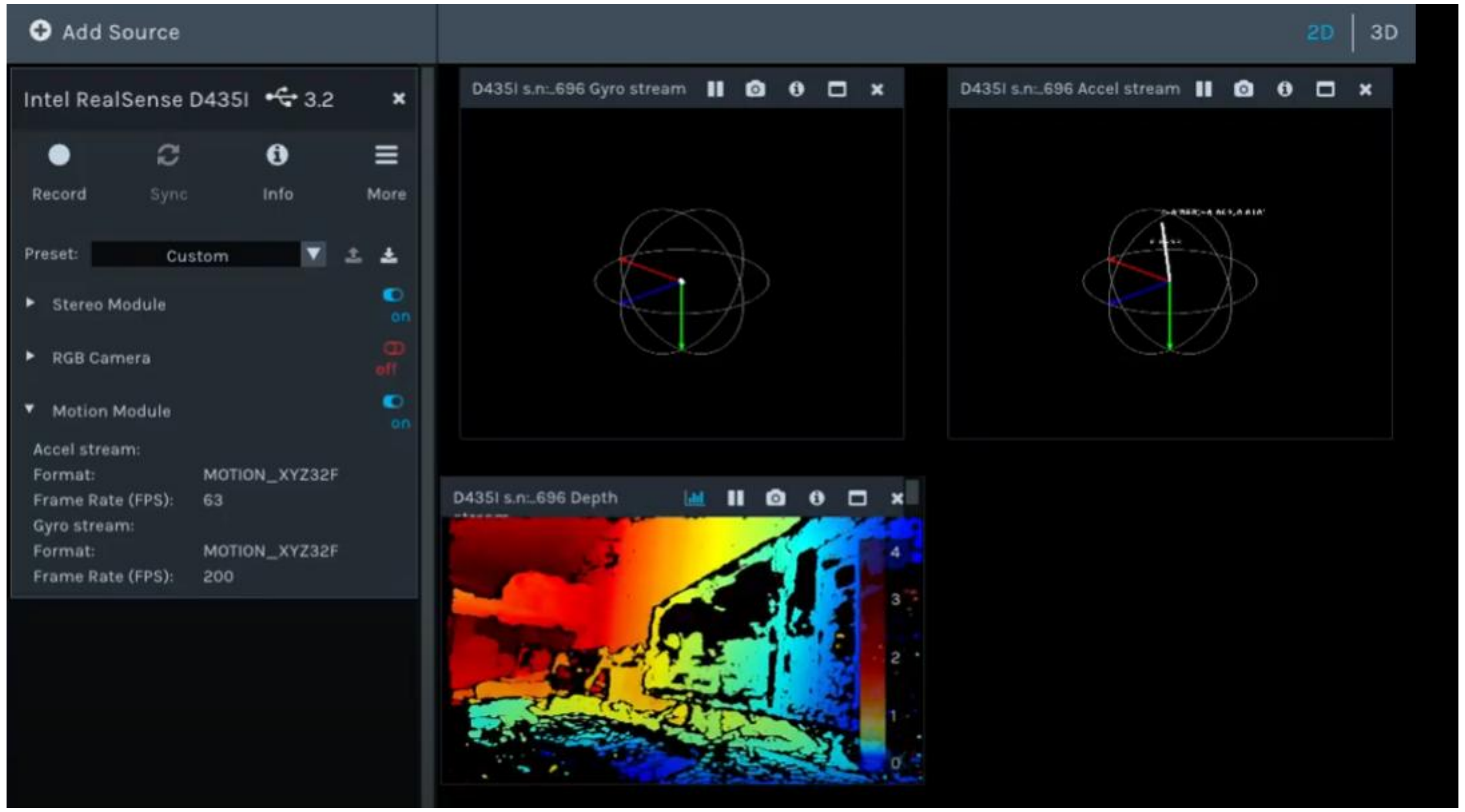
1. Jira를 활용한 프로젝트 이슈 관리

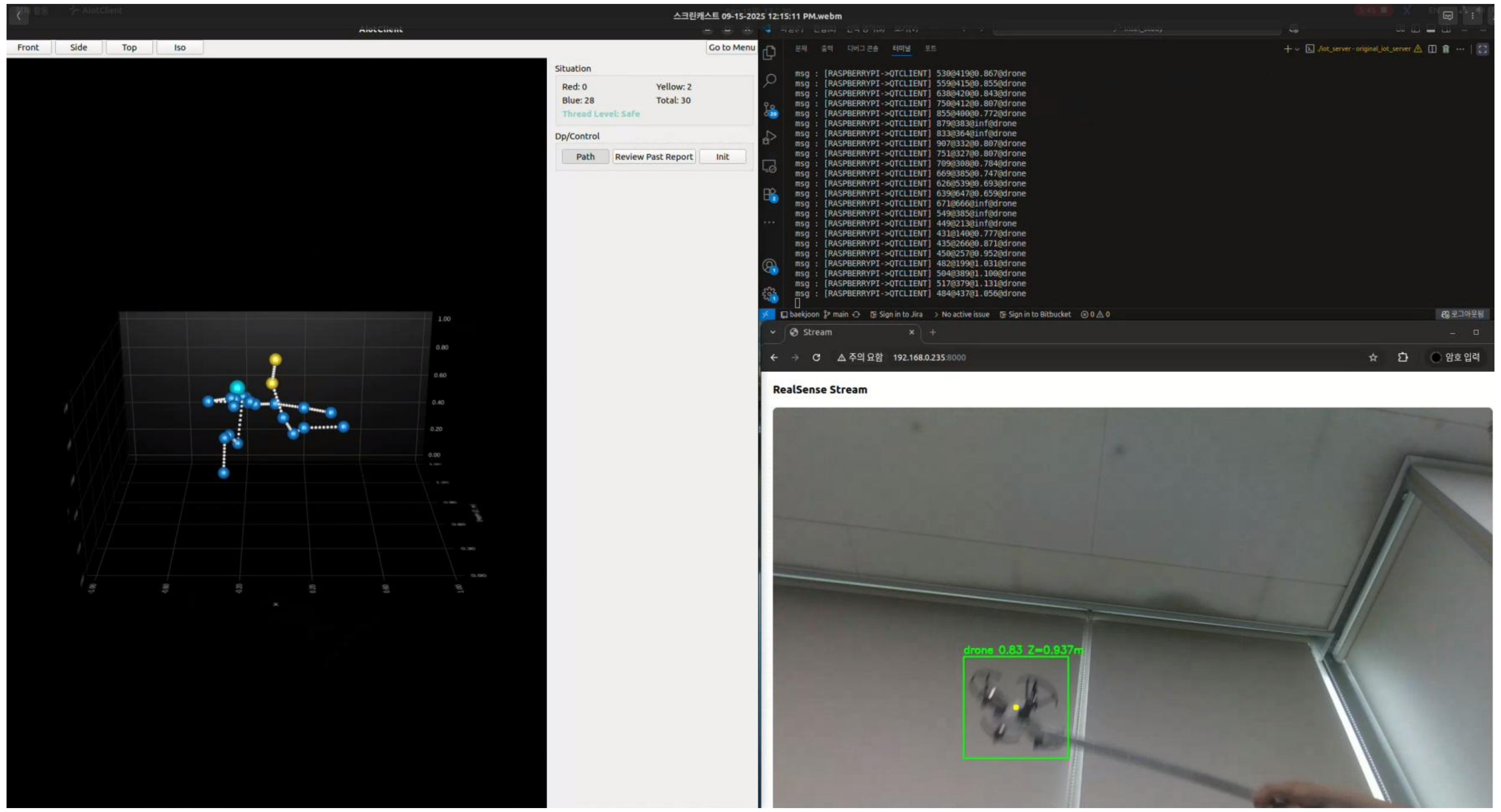




05 이슈

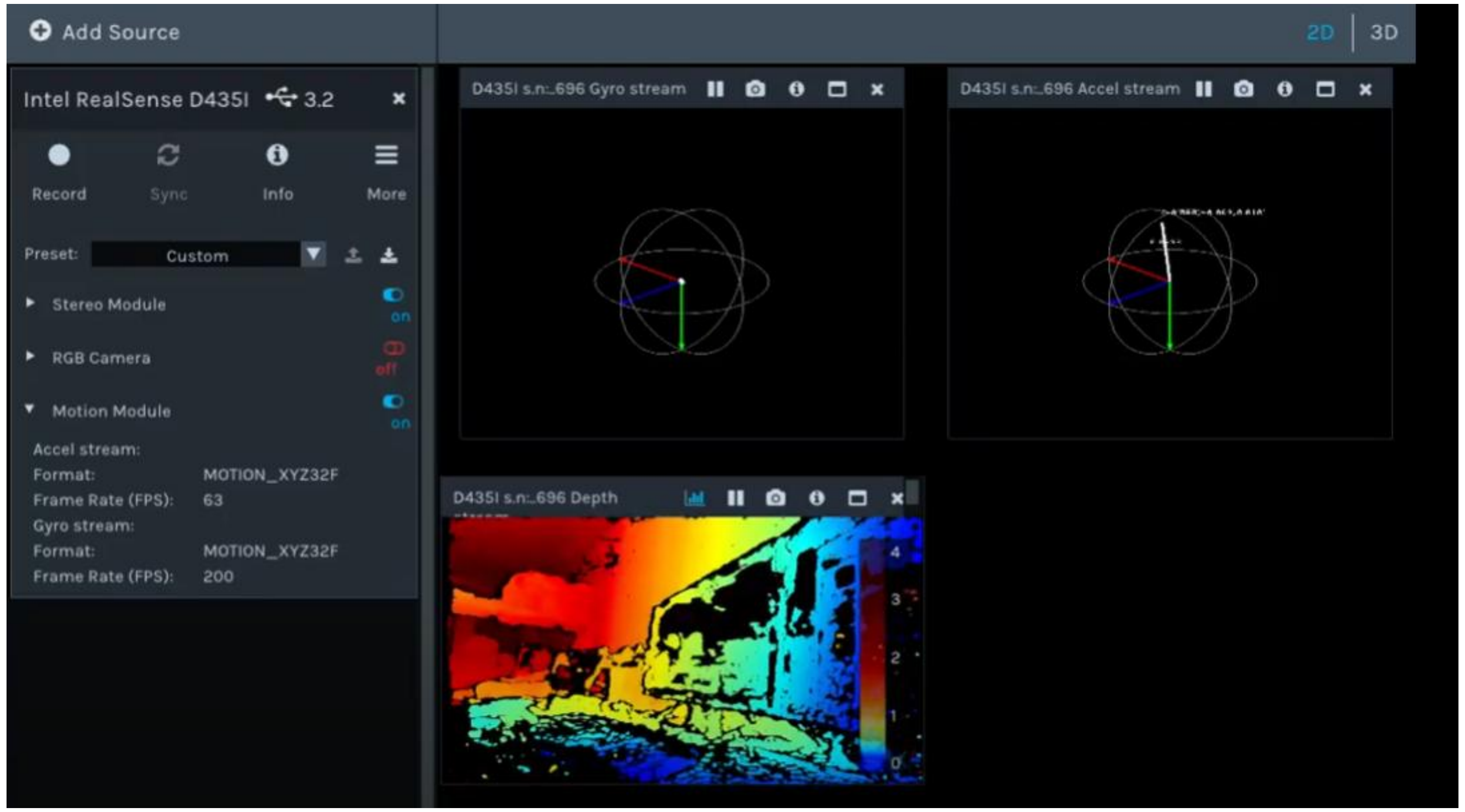
2. Object의 3D 좌표 정의

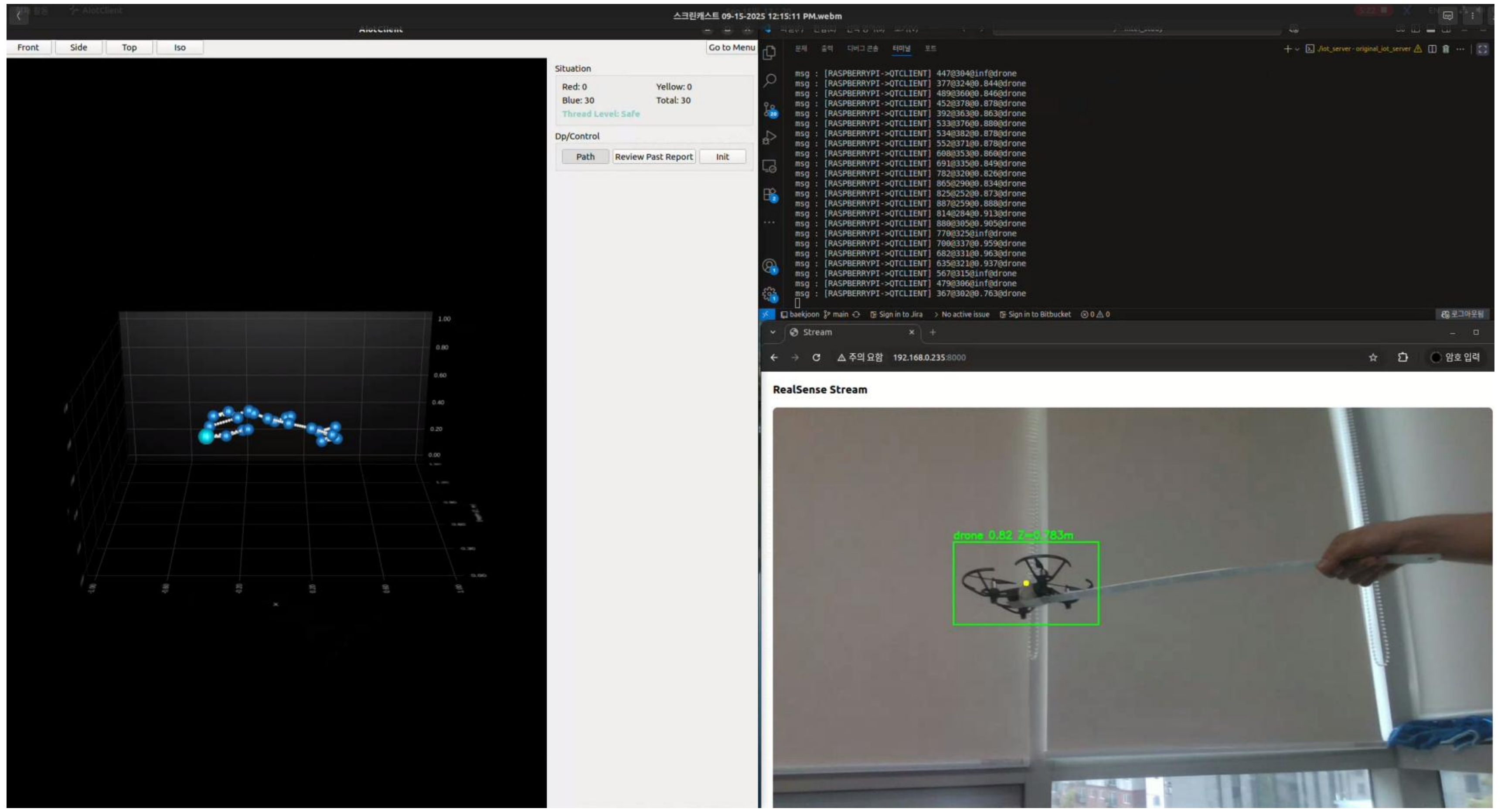




05 이슈

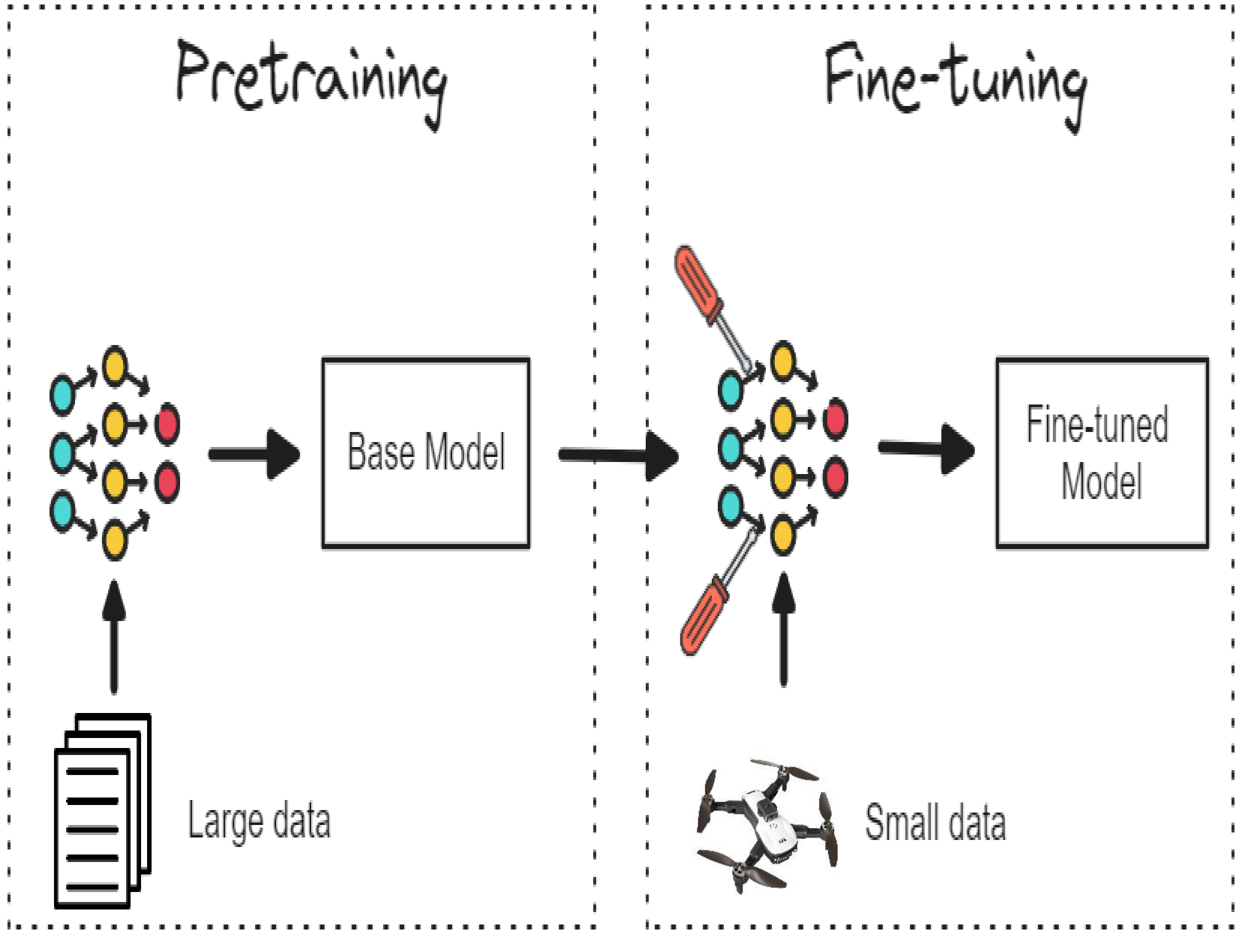
2. Object의 3D 좌표 정의



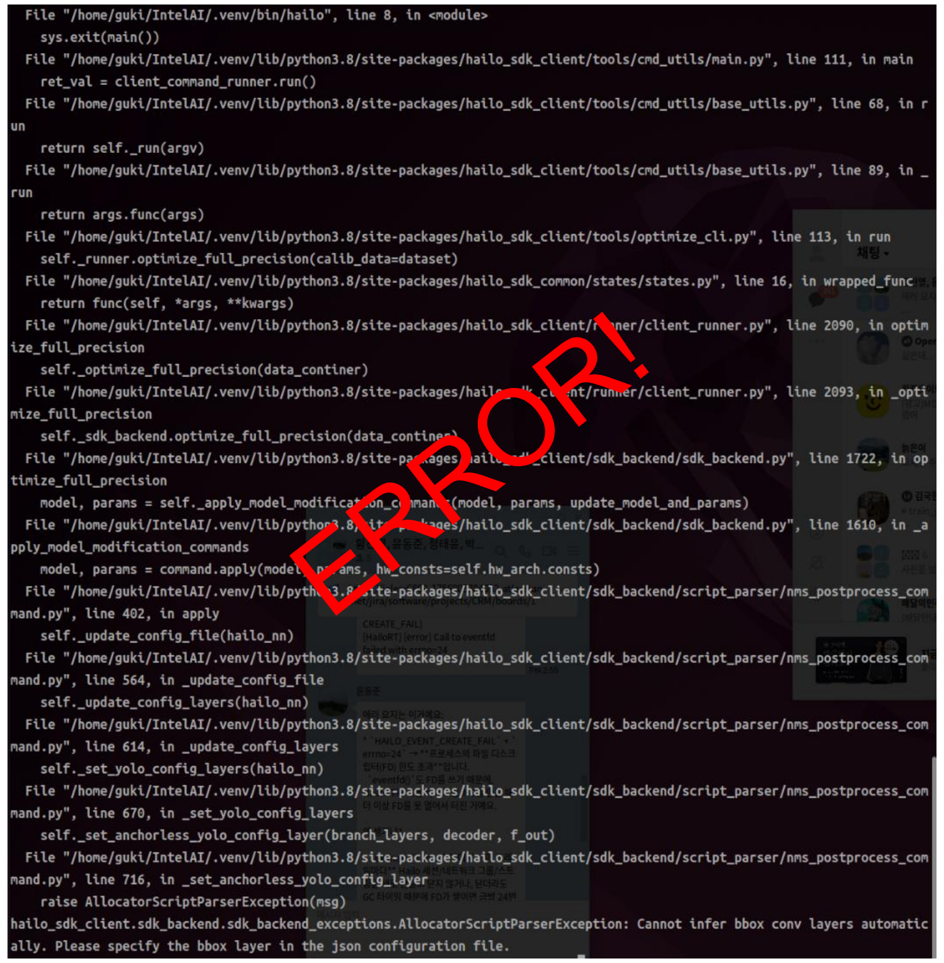


05 이슈

3. Hailo 관련 이슈



Compile

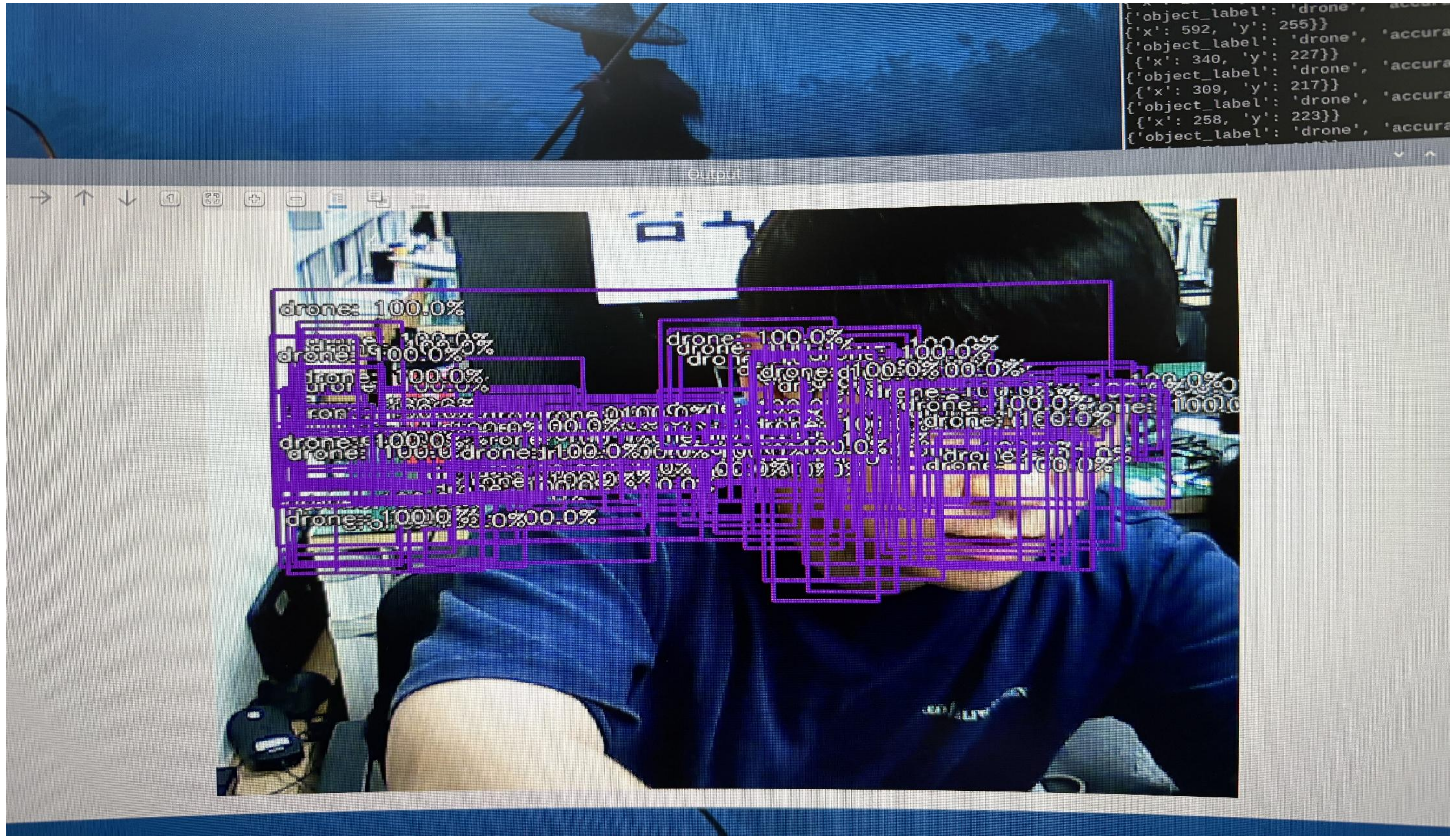


05 이슈

3. Hailo 관련 이슈

Application에서 후처리 진행

But 설정 파일 parameter matching 오류, 후처리 오류



05 이슈

3. Hailo 관련 이슈

Question

기본적으로 저희 model conversion 시에 사용되는 정보들과 DFC(DataFlow Compiler) configuration 등은 Model zoo github(GitHub - hailo-ai/hailo_model_zoo: The Hailo Model Zoo includes pre-trained models and a full building and evaluation environment)에서 예시들을 공유하고 있습니다.

yolov8을 기반으로
hailo_model_zoo/hailo_model_zoo/cfg/networks/yolov8s.yaml at master · hailo-ai/hailo_model_zoo · GitHub

```
base:
- base/yolov8.yaml
postprocessing:
  device_pre_post_layers:
    nms: true
  hpp: true
network:
  network_name: yolov8s
paths:
  network_path:
  - models_files/ObjectDetection/Detection-COCO/yolo/yolov8s/2023-02-02/yolov8s.onnx
  alls_script: yolov8s.alls
  url: https://hailo-model-zoo.s3.eu-west-2.amazonaws.com/ObjectDetection/Detection-COCO/yolo/yolov8s/2023-02-02/yolov8s.zip
info:
  task: object detection
  input_shape: 640x640x3
  output_shape: 80x5x100
  operations: 28.6G
  parameters: 11.2M
  framework: pytorch
  training_data: coco train2017
  validation_data: coco val2017
  eval_metric: mAP
  full_precision_result: 44.58
  source: https://github.com/ultralytics/ultralytics
  license_url: https://github.com/ultralytics/ultralytics/blob/main/LICENSE
  license_name: AGPL-3.0
```

.yaml 파일은 input node 부터 nms가 모델 내부에 포함이 되어있다(nms: true) 등의 정보들을 포함한 내용이고,

hailo_model_zoo/hailo_model_zoo/cfg/alls/generic/yolov8s.alls at master · hailo-ai/hailo_model_zoo · GitHub
.alls 파일은 DFC로 모델 변환 시에 들어가는 옵션들이 나열되어 있습니다.

hailo_model_zoo / hailo_model_zoo / cfg / alls / generic / yolov8s.alls

HailoModelZoo update-to-version-2.11.0

Code Blame 5 lines (5 loc) · 295 Bytes

```
normalization1 = normalization([0.0, 0.0, 0.0], [255.0, 255.0, 255.0])
change_output_activation(conv42, sigmoid)
change_output_activation(conv53, sigmoid)
change_output_activation(conv63, sigmoid)
nms_postprocess("../../postprocess_config/yolov8s_nms_config.json", meta_arch=yolov8, engine=cpu)
```

https://mail.naver.com/v2/popup/read/0/8474/print 1/4

25. 9. 15. 오후 7:29 Fwd: hailo 문의

위 자료를 참고하시어 모델 변환 하시는 것을 보통은 reference로 잡고 계시며,
변환 코드는 DFC를 설치하셨을 때,
<venv> /lib64/python3.8/site-packages/hailo_tutorials/notebooks/*.ipynb
파일들을 보시면, 1_Parsing tutorial -> 2_Model_Optimization -> 3_Compilation
등의 튜토리얼이 모델 변환 과정입니다. 코드를 참조하시고,

모델 변환 시에 NMS를 모델 파일에 같이 넣어서 실행시킬 것인가, Post-Processing을 CPU(in application code)에서 하실 것인가를 설정할 수 있는데,
위와 같이 NMS를 넣을 때는 그에 맞는 yolov8s_nms_config.json 파일이 필요합니다.
<venv>/lib/python3.8/site-packages/hailo_sdk_client/tools/core_postprocess/default_nms_config_yolov8.json
위 파일을 참조하시면, 모델에서 Parsing 시에 변환된 이름으로 위와 같은 포맷을 넣어야 NMS를 넣고 빼는게 가능해지십니다.

Feedback

&

05 이슈

3. Hailo 이슈 해결

05 이슈

4. 라즈베리파이 브로드컴 성능 문제

1. 통신 속도 저하
2. 영상 스트리밍 중단
3. 트래킹 연산 속도 저하

06 향후 계획

1. 더 경량화된 드론 추적 시스템을 완성할 수 있는 CPU를 사용

2. 라벨 추가를 통해 헬기, 비행기 등 더 다양한 분류구현

3. 다양한 AI 모델을 테스트하여 경량화된 드론 추적에 적합한 AI 모델 최적화

감사합니다

THANK YOU!